Lab Report 01

Course: Signals and Systems Lab (ICT-2107) **Experiment No.:** 01 **Experiment Name:** Generation and Plotting of Continuous-Time Signals **Name:** Sobuj **Roll / ID:** _____ **Date of Experiment:** _____ **Date of Submission:** _____

1. Objective

The objectives of this experiment are:

- To generate continuous-time (CT) signals using a computer program (MATLAB/Octave).
- To understand how a CT signal is approximated by sampling it on a very fine time grid.
- To generate a cosine signal, a scaled and phase-shifted cosine signal over a specified number of periods, and a real exponential signal.
- To plot all the signals in a single figure using subplots and compare their behaviour.

2. Problem Statement

Write a program to generate the following continuous-time signals and plot all of them in a single figure:

1. $x(t) = \cos(t)$, for $0 \leqslant t \leqslant 10$
2. $y(t) = 3\cos(3\pi t + \pi/3)$, for four periods
3. $y(t) = 2e^{(0.9t)}$, for $0 \leqslant t \leqslant 10$

3. Theory

3.1 Continuous-time signals

A continuous-time signal is defined for every value of the independent variable t. A computer cannot store infinitely many points, so a CT signal is represented by evaluating it at a very small time step (for example 0.01 s). When the step is small enough, the plotted curve looks smooth and continuous.

3.2 Sinusoidal (cosine) signal

The general form of a cosine signal is

$x(t) = A\cos(\omega t + \varphi)$

where A is the amplitude, $\omega = 2\pi f$ is the angular frequency in rad/s, and φ is the phase in radians. The fundamental period is

$T = 2\pi/\omega = 1/f$

Signal (i): $x(t) = \cos(t)$. Here $A = 1$, $\omega = 1$ rad/s, $\varphi = 0$, so $T = 2\pi \approx 6.283$ s. Over $0 \leqslant t \leqslant 10$ the signal completes about 1.59 cycles.

Signal (ii): $y(t) = 3\cos(3\pi t + \pi/3)$. Here $A = 3$, $\omega = 3\pi$ rad/s, $\varphi = \pi/3$ rad (60°), so

$f = \omega/2\pi = 1.5$ Hz, $T = 2\pi/(3\pi) = 2/3$ s ≈ 0.667 s

Four periods therefore correspond to the time span $4T = 8/3$ s ≈ 2.667 s, so the time axis is taken as $0 \leqslant t \leqslant 4T$. The phase shift moves the waveform to the left by $\varphi/\omega = (\pi/3)/(3\pi) = 1/9$ s, so at $t = 0$ the value is $3\cos(\pi/3) = 1.5$.

3.3 Real exponential signal

A real exponential signal has the form

$y(t) = C\,e^{(at)}$

If $a > 0$ the signal grows exponentially, and if $a < 0$ it decays exponentially. For **Signal (iii):** $y(t) = 2e^{(0.9t)}$, $C = 2$ and $a = 0.9 > 0$, so the signal is a growing exponential. It starts at $y(0) = 2$ and reaches $y(10) = 2e^{9} \approx 16{,}206$ at $t = 10$.

Note: If the exponent in the lab sheet is negative ($2e^{(-0.9t)}$), only the sign of the exponent in the code changes, and the signal becomes a decaying exponential from 2 to about 2.2×10^{-4}.

4. Apparatus

- A computer with MATLAB (or GNU Octave)
- Script editor and command window

5. Algorithm

1. Clear the workspace and close all figures.
2. Define the time vector t1 from 0 to 10 with a small step (0.01) for signals (i) and (iii).
3. Compute the period T = 2/3 of signal (ii), and define the time vector t2 from 0 to 4T.
4. Evaluate the three signals using element-wise operations.
5. Create one figure with three subplots, and add title, axis labels and grid to each.
6. Run the program and observe the output.

6. MATLAB Code

```
clc; clear; close all;

%% Time vectors
dt = 0.01;
t1 = 0:dt:10;              % for signals (i) and (iii)

w  = 3*pi;                 % angular frequency of signal (ii)
T  = 2*pi/w;               % period = 2/3 s
t2 = 0:dt/10:4*T;          % four periods

%% Signals
x1 = cos(t1);                    % (i)   cos(t)
y2 = 3*cos(3*pi*t2 + pi/3);      % (ii)  3cos(3*pi*t + pi/3)
y3 = 2*exp(0.9*t1);              % (iii) 2e^(0.9t)

%% Plot all signals in a single figure
figure;

subplot(3,1,1);
plot(t1, x1, 'b', 'LineWidth', 1.5);
title('(i) x(t) = cos(t), 0 \leq t \leq 10');
xlabel('Time, t (s)'); ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t2, y2, 'r', 'LineWidth', 1.5);
title('(ii) y(t) = 3cos(3\pi t + \pi/3), four periods');
xlabel('Time, t (s)'); ylabel('Amplitude');
grid on;

subplot(3,1,3);
plot(t1, y3, 'k', 'LineWidth', 1.5);
title('(iii) y(t) = 2e^{0.9t}, 0 \leq t \leq 10');
xlabel('Time, t (s)'); ylabel('Amplitude');
grid on;
```

7. Results and Observations

(Insert the screenshot of the output figure here.)

Signal	Expression	Amplitude	Period	Time range	Nature
(i)	cos(t)	1	2π ≈ 6.283 s	0 to 10 s	Periodic, oscillating

Signal	Expression	Amplitude	Period	Time range	Nature
(ii)	3cos(3πt + π/3)	3	2/3 ≈ 0.667 s	0 to 8/3 s	Periodic, phase-shifted
(iii)	2e^(0.9t)	2 at t = 0	Not periodic	0 to 10 s	Growing exponential

Observations:

- Signal (i) oscillates between +1 and −1, starts at its maximum value 1 at t = 0, and completes one full cycle at t ≈ 6.28 s.
- Signal (ii) oscillates between +3 and −3 and exactly four complete cycles are visible in the plotted interval. At t = 0 its value is 1.5 because of the π/3 phase shift, and its first peak occurs at t = 1/9 s earlier than a pure cosine would, so the curve appears shifted to the left.
- Signal (iii) is not periodic. It grows rapidly, starting from 2 and reaching about 1.62×10^4 at t = 10 s.

8. Discussion

In this experiment the continuous-time signals were generated by sampling them at a very small time step and joining the points with straight lines, which gives a smooth curve that closely approximates the true CT signal. Element-wise operations were used so that the whole signal was calculated in one step instead of using loops.

For signal (ii) the time axis was chosen as a multiple of the period (4T) instead of a fixed range, so that exactly four periods are displayed. The effect of amplitude scaling (A = 3) and phase shift (φ = π/3) on a cosine wave was clearly observed when compared with the plain cos(t).

Signal (iii) shows that exponential signals have a very different nature from sinusoids: they are not periodic and they either grow or decay depending on the sign of the exponent. Because the growth is very large, the early part of the curve looks almost flat when plotted on a linear scale.

The use of `subplot` allowed all the signals to be shown in a single figure, which makes comparison easy. Care must also be taken that the time step is small enough, otherwise the high-frequency signal (ii) would look distorted.

9. Conclusion

The three continuous-time signals, a cosine, a phase-shifted and scaled cosine over four periods, and a real exponential, were successfully generated and plotted in a single figure using MATLAB. The plotted amplitude, period and phase of each signal matched the

values obtained from the theoretical calculations, so the objectives of the experiment were achieved.